A054 v0.6.0: E013 Provider-Convergent State-Selection Falsifier

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1 Identification

This v0.6.0 software migration preserves the parent scientific hypotheses and numerical thresholds. It does not execute state selection, Hessian, ringdown, native parity or reveal. The public migration protocol is frozen separately from any future complete blind campaign.

Catalog ID A054; release v0.6.0; SST Falsifier Framework v1.0.10, registered profile `multilibrary_gpu`; this execution uses only the bounded ADMISSION_SMOKE lane. This is an E013 cross-falsifier member at stage 30.

2 Question

Can distinct current E011/PKLSA provider representatives of the same topology be mapped by one target-free variational selector to a common stationary and locally stable finite-core state?

3 Hypotheses

The null states that provider dependence survives target-free state selection or that selected stationary states fail local stability. The alternative requires provider-convergent stationary and stable endpoints without topology/provider-specific retuning.

4 Assumptions

E013 fixes the current carrier population. The selector is a variational existence/selection test, not a claim that inviscid Euler flow is dissipative. Physical stability is tested separately after stationarity. Legacy falsifier results are excluded as evidence.

5 Symbols

All hatted lengths, deformation coordinates and the comparison Hamiltonian are dimensionless under the registered reach normalization. No calibrated SI length, time, energy or physical dissipation is inferred.

$\widehat{H}_a$ is the reach-normalized regularized line-energy kernel, q Bishop-frame deformation coordinates, σ the normalized energy-gradient stationarity score, and K_H the endpoint Hessian.

6 Equations

Machine-readable science contract

Research question On one E013-frozen population of current E011 provider anchors, can a fixed target-free variational state-selection procedure map distinct source representatives of the same topology to a common stationary and locally stable finite-core state, without using legacy falsifier results?

Objective Re-run state selection from source-native bytes for exactly the E013 carriers using one canonical constrained objective Uc(q)=H(N[R(X+deltaX(q))]) for gradient, Armijo line search, Hessian and ringdown; quantify stationarity, topology preservation, provider-endpoint convergence and local stability; numerical inability to demonstrate descent or Hessian-sign convergence is UNRESOLVED rather than a scientific FAIL; freeze only provider-converged stable endpoints as a future A057 handoff.

H_0 H0: provider-distinct current PKLSA/SKLSA representatives do not converge under the preregistered selector to a reproducible stationary and stable topology-level state beyond numerical/source variation.

H_1 H1: the same preregistered selector, without topology/provider-specific retuning, yields provider-convergent stationary and locally stable endpoints for a reproducible subset of topologies.

Registered equations

E1: Regularized finite-core comparison Hamiltonian

$$\widehat{H}_a = \sum_{i,j} \oint \oint \frac{d\widehat{\mathbf{X}}_i \cdot d\widehat{\mathbf{X}}_j}{\sqrt{\|\widehat{\mathbf{X}}_i - \widehat{\mathbf{X}}_j\|^2 + \widehat{a}^2}} \tag{1}$$

Dimensional check: All lengths are reach-normalized; the kernel is dimensionless.

E2: Bishop-Fourier reduced deformation

$$\delta\mathbf{X}(q) = \sum_{\mu} q_\mu \mathbf{B}_\mu, \qquad \mathbf{B}_\mu \perp \mathbf{t} \tag{2}$$

Dimensional check: In reach-normalized coordinates both q and B are dimensionless.

E3: Canonical constrained objective and stationarity score

$$\mathcal{P} = \mathcal{N} \circ \mathcal{R}, \qquad U_c(q; X) = \widehat{H}_a(\mathcal{P}[X + \delta X(q)]), \qquad \sigma = \frac{\|\nabla_q U_c(0; X)\|_2}{\max(|U_c(0; X)|, \epsilon)} \tag{3}$$

Dimensional check: Projection and the reach-normalized Hamiltonian are dimensionless, so U_c and sigma are dimensionless.

E4: Target-free Armijo/trust-region selector on the constrained objective

$$d_n = -\frac{\nabla U_c}{\|\nabla U_c\|}, \qquad \eta_{n,k} = \min(\Delta_n, \eta_0)\beta^k, \qquad U_c(\eta_{n,k} d_n) \le U_c(0) + c_1 \eta_{n,k} \nabla U_c \cdot d_n \tag{4}$$

Dimensional check: q, trust radius and U_c are dimensionless in the reach-normalized lane.

E5: Accepted-step integrity conditions

$$\widehat{H}_{n+1} \le \widehat{H}_n, \quad d_{\min,n+1} \ge f_c d_{\min,0}, \quad |\Delta Lk| \le \varepsilon_{Lk} \tag{5}$$

Dimensional check: Energy comparison is dimensionless; distance is compared as a ratio; linking is dimensionless.

E6: Cross-provider endpoint agreement

$$CV_H(K) = \frac{s(H_K^*)}{|\overline{H}_K^*|}, \qquad \Delta_R(K) = \frac{R_{\max}^* - R_{\min}^*}{|\overline{R}^*|} \tag{6}$$

Dimensional check: Coefficient of variation and relative spans are dimensionless.

E7: Constrained-objective Hessian with finite-difference sign convergence

$$K_H^{(j)} = \nabla_q^2 U_c(0; \epsilon/2^j), \quad j = 0, 1, 2; \qquad \text{promote sign} \iff s_0 = s_1 = s_2 \ \wedge \ \frac{\|K_H^{(2)} - K_H^{(1)}\|_F}{\max(\|K_H^{(2)}\|_F, \epsilon)} \le \tau_H \tag{7}$$

Dimensional check: The reduced coordinates and U_c are dimensionless, so all Hessian entries/eigenvalues are dimensionless.

E8: Nonlinear reduced ringdown on the same constrained objective

$$\ddot{z} = -\partial_z U_c(z\, e_{\min}), \qquad e_{\min} = \operatorname*{argmin}_{\|e\|=1} e^T K_H e \tag{8}$$

Dimensional check: Normalized time, z and U_c make the reduced test dimensionless.

E9: State-selected handoff predicate

$$Q_K = I[\text{provider consensus}] I[\sigma \le \sigma_{\max}] I[K_H \succeq 0] I[\text{ringdown bounded}] \tag{9}$$

Dimensional check: Boolean predicate.

Registered code-to-mathematics steps

1. **S1** [G0; equations E1]: Verify frozen protocol/implementation and exact E013/E011 source-byte binding.
2. **S2** [G1; equations E1]: Reload only E013-frozen carriers and recheck geometry hashes.
3. **S3** [G2; equations E1, E3, E4, E5]: Validate rigid-energy invariance, exact constrained-objective evaluation, Armijo monotonicity and topology-proxy mechanics.
4. **S4** [G3; equations E2, E3, E4, E5]: Run the fixed constrained-objective variational selector independently on every current carrier; classify failure to demonstrate numerical descent as UNRESOLVED_NUMERICAL.
5. **S5** [G4; equations E3, E4]: Evaluate fixed-selector transfer on registered confirmation topologies.
6. **S6** [G5; equations E1]: Certify the common energy kernel against C++/OpenMP FP64.
7. **S7** [G6; equations E1]: Record optional GPU/DD32 framework screening without promotion.
8. **S8** [G7; equations E6]: Compare independently sourced endpoints topology-by-topology.
9. **S9** [G8; equations E7, E8]: Qualify stationary endpoints only after constrained-Hessian eps/2/4 convergence and then nonlinear ringdown on the same U_c.

10. **S10** [G10; equations E6]: Test whether initially provider-sensitive energy kernels collapse after selection.

11. **S11** [G11; equations E5]: Audit clearance/linking topology proxies over all paths.

12. **S12** [G12; equations E9]: Freeze only provider-converged stable endpoints as a future A057 handoff.

13. **S13** [G22; equations E3, E6, E7, E9]: Assign the blind overall state-selection classification under diagnostic continuation.

14. **S14** [G9; equations E9]: Reserve framework reveal integrity gate; v0.6.0 contains no mass target reveal.

7 Algorithm

The registered full algorithm reloads carriers and evaluates one constrained objective $U_c(q) = \widehat{H}_a(\mathcal{N}[\mathcal{R}(X + \delta X(q))])$. The same U_c supplies finite-difference gradients, Armijo/trust-region acceptance, stationarity, endpoint Hessians, and nonlinear reduced ringdown. Accepted steps must preserve a nonlocal-clearance floor and pairwise linking within tolerance. Failure to demonstrate numerical descent is recorded as `UNRESOLVED_NUMERICAL`, not as negative physical evidence.

8 Sources

E011-v0.3.0 is the static-seed/provider authority; E013-v0.1.1 freezes the common carrier population; E010-v0.3.1 only reloads the exact source-native geometry bytes. No old A054 output is a science input.

9 Gates and diagnostic continuation

Only G0 is a hard prerequisite. Scientific failures do not suppress later diagnostics. Soft dependency caveats prevent a numerical diagnostic from being promoted beyond its support.

10 Numerics

BASIC, FULL and CERTIFY increase reduced basis richness, resolution, iteration budget and stationarity strictness. The same global Armijo/trust-region policy is used for every carrier; no provider/topology-specific optimization hyperparameter is fitted. Endpoint Hessians are checked at ϵ, $\epsilon/2$ and $\epsilon/4$ and their sign is promoted only after finite-difference convergence.

11 Backends

Python/NumPy FP64 is REFERENCE. The registered C++/pybind11 OpenMP FP64 parity criterion for the common line-energy kernel is relative-L2 $\leq 10^{-10}$. Framework SYCL/DD32 remains SCREENING_ONLY. Neither native parity nor GPU screening was executed in this admission smoke.

12 Blindness and provenance

There are no protected mass targets in v0.6.0. Blindness is used to lock the protocol and to exclude legacy scientific outputs. E013 carrier identities and provider groups remain visible because provider convergence is the object of the test.

13 Statistics

Primary statistics are stationarity success lower/upper bounds under explicit numerical-unresolved accounting, provider endpoint energy CV, ropelength relative span, absolute-writhe span, source-to-endpoint provider-CV collapse, Hessian finite-difference convergence/sign and nonlinear ringdown amplitude/energy drift.

14 Results

Controlled migration scope

This run only verifies source admission and reloads exact geometry. State selection, native parity, Hessian, ringdown and reveal were not executed. A software PASS is not scientific certification.

Axis	Status
migration_gate	PASS_SOURCE_AND_ADAPTER
execution_gate	PASS_BOUNDED_ADMISSION_SMOKE
scientific_gate	NOT_RUN
scientific_status_promoted	False

Exact admitted carrier reloads

Topology	Provider	Static seed	Points
3_1	gilbert	SEED_25f0195c1c823d27	4096
5_2	gilbert	SEED_ff6a64029ad73887	4096
6_1	gilbert	SEED_b54595f817130fbb	4096
4_1	gilbert	SEED_b004004933759224	4096
L2a1	gilbert	SEED_92e4935a329b47db	8192

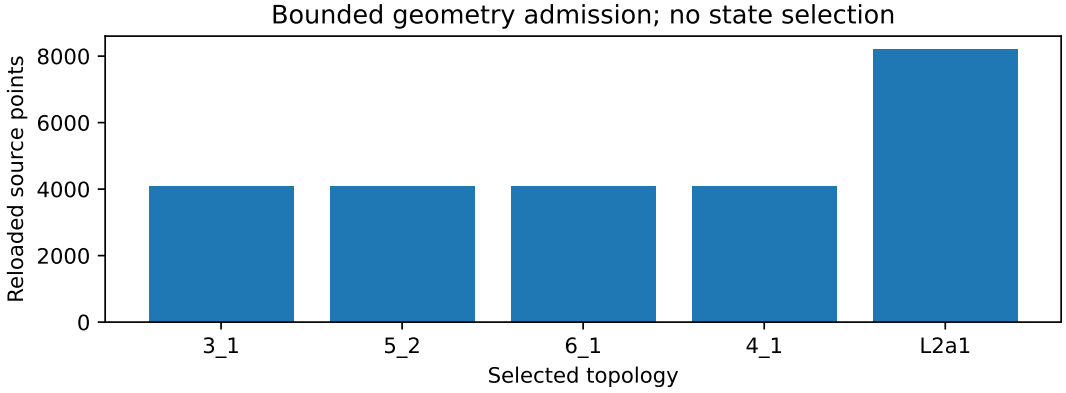


Figure 1: Actual source-native point counts reloaded in this admission smoke. Counts are not a convergence or physical-quality score.

Unresolved scientific evidence

Each BASIC topology has one selected provider. Independent provider convergence cannot close, including L2a1. No stationary endpoint, converged Hessian, ringdown or state-selected handoff was computed. Existing scientific thresholds remain unchanged.

Automatic run provenance

Field	Recorded value
Run ID	20261026T222157498479Z-0ee220b5
Run phase	BLIND
Protocol bundle SHA-256 (byte exact)	e2a68d01ad0622700daaad7aed8151dd34d36ad76ea4c37d e5492e5dd9120ab8
Portable protocol diagnostic SHA-256	unavailable
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Science evidence SHA-256	6625dc91633dfe38045f30676871a21fdf17877542be22b0 bd8786c2951ef412
Final output manifest	See OUTPUT_MANIFEST.json (finalized after PDF and report metadata)

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Is the frozen v0.6.0 protocol intact and runtime/implementation provenance self-consistent?	112 qualified framework files and public scoped migration protocol verified; not a complete physical blind FREEZE.
G1	PASS	Is the run exactly bound to the E013 common-carrier manifest, with all E011 source geometries reloaded and hash-verified?	Fresh E013 selector binding; exact current E011 joins, raw hashes and parsed geometry hashes verified.
G2	UNRESOLVED	Do rigid-objectivity, constrained-objective identity, Armijo monotonicity and topology-proxy controls pass?	Registered scientific evaluator not executed in the bounded admission lane; no scientific result is asserted.
G3	UNRESOLVED	Can the fixed target-free constrained-objective selector drive the preregistered carrier fraction to stationary states, with numerical descent failures retained as UNRESOLVED?	Registered scientific evaluator not executed in the bounded admission lane; no scientific result is asserted.
G4	UNRESOLVED	Does the same frozen constrained-objective selector generalize without refit to registered confirmation topologies?	Registered scientific evaluator not executed in the bounded admission lane; no scientific result is asserted.
G5	UNRESOLVED	Does C++/OpenMP FP64 reproduce Python/NumPy FP64 for the state-selection energy kernel?	Registered scientific evaluator not executed in the bounded admission lane; no scientific result is asserted.
G6	UNRESOLVED	If available, does framework GPU/DD32 screening evidence remain consistent without being promoted to certification?	Registered scientific evaluator not executed in the bounded admission lane; no scientific result is asserted.

G7	UNRESOLVED	Do independently sourced provider anchors of the same topology converge to the same stationary endpoint without unresolved optimizer cases?	Registered scientific evaluator not executed in the bounded admission lane; no scientific result is asserted.
G8	UNRESOLVED	Are stationarity-qualified endpoints locally stable under a finite-difference-converged constrained Hessian and bounded ringdown on the same objective?	Registered scientific evaluator not executed in the bounded admission lane; no scientific result is asserted.
G10	UNRESOLVED	For source-energy-sensitive topologies, does numerically resolved state selection collapse provider spread toward a common endpoint?	Registered scientific evaluator not executed in the bounded admission lane; no scientific result is asserted.
G11	UNRESOLVED	Are clearance and link-drift topology proxies preserved along every completed state-selection path?	Registered scientific evaluator not executed in the bounded admission lane; no scientific result is asserted.
G12	UNRESOLVED	Can at least one provider-converged stationary, Hessian-converged, stable state be frozen as an A057-compatible handoff?	Registered scientific evaluator not executed in the bounded admission lane; no scientific result is asserted.
G22	UNRESOLVED	What is the overall state-selection evidence classification when numerical unresolved states are kept distinct from scientific FAIL?	Registered scientific evaluator not executed in the bounded admission lane; no scientific result is asserted.
G9	DEFERRED	After a completed FULL/CERTIFY blind run, do reveal and commitment checks verify exactly?	Reveal is outside the authorized admission-smoke scope.

Scientific qualification (separate from measured gate outcomes)

Gate	Evidence qualification	Blocking prerequisites
G0	QUALIFIED	none
G1	QUALIFIED	none
G2	UNRESOLVED	none
G3	UNRESOLVED	none
G4	UNRESOLVED	none
G5	UNRESOLVED	none
G6	UNRESOLVED	none
G7	UNRESOLVED	none
G8	UNRESOLVED	none
G10	UNRESOLVED	none
G11	UNRESOLVED	none
G12	UNRESOLVED	none
G22	UNRESOLVED	none
G9	DEFERRED	none

Backend manifest


```
{
  "actual_backend": "Python/NumPy FP64 source parsing",
  "authority": "SOURCE_ADMISSION_ONLY",
  "gpu_used": false,
  "native_executed": false,
  "physical_qualification": "NOT_RUN",
```

```
  "schema": "SST-BACKEND-MANIFEST-2"
}
```


Source manifest


```
{
  "manifest_sha256":
"0b4f20bfec153f5179b07f2c573009688b7a75e23e03a1deea843568df47d4ad",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "E011",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "gilbert",
      "path": "C:\\workspace\\projects\\SST-Workbench\\03_data\\A_knots\\01_i
deal\\ideal_sources\\Ideal.txt.gz",
      "provenance_family": null,
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "Admitted geometry; static authority only",
      "sha256":
"274a4ce111607a6295e280f3f8369c665e7de82ac33afcd4cb227eb61568d530",
      "source_id": "CAR_f0023c079d7502c1",
      "version": "v0.3.0"
    },
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      "exists": true,
      "independence_family": "gilbert",
      "path": "C:\\workspace\\projects\\SST-Workbench\\03_data\\A_knots\\01_i
deal\\ideal_sources\\Ideal.txt.gz",
      "provenance_family": null,
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
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      "sha256":
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      "source_id": "CAR_97c30a73f3ab9b85",
      "version": "v0.3.0"
    },
    {
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      "independence_family": "gilbert",
      "path": "C:\\workspace\\projects\\SST-Workbench\\03_data\\A_knots\\01_i
deal\\ideal_sources\\Ideal.txt.gz",
      "provenance_family": null,
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "Admitted geometry; static authority only",
      "sha256":
"274a4ce111607a6295e280f3f8369c665e7de82ac33afcd4cb227eb61568d530",
      "source_id": "CAR_154a6dad7b515c61",
      "version": "v0.3.0"
    },
    {
```

```
      "catalog_id": "E011",
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      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "Admitted geometry; static authority only",
      "sha256":
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      "source_id": "CAR_fc293690c619f8ca",
      "version": "v0.3.0"
    },
    {
      "catalog_id": "E011",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "gilbert",
      "path": "C:\\workspace\\projects\\SST-Workbench\\03_data\\A_knots\\01_i
deal\\ideal_sources\\IdealLinks.txt.gz",
      "provenance_family": null,
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "Admitted geometry; static authority only",
      "sha256":
"fac7791b94819e14718204dc0b5ff2e15fcc0a71e2b6da7b4bd1a8f5a32184be",
      "source_id": "CAR_a3ffb7da038a2929",
      "version": "v0.3.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Interpretation

A stationary point is not a stable state merely because $\nabla U_c \simeq 0$. Provider convergence and positive local stability are independent requirements. Numerical inability to establish descent or Hessian-sign convergence remains unresolved. Only topologies satisfying provider convergence, stationarity, converged nonnegative local stability and bounded ringdown are eligible for the state-selected handoff.

16 Reproducibility

From this version directory run `run_all.cmd SELFTEST` or `run_all.cmd ADMISSION_SMOKE`. CPython 3.14, NumPy, matplotlib, pytest and pdflatex are required, together with this Workbench's exact pinned framework, E010 parser and E011 sources. FULL, CERTIFY and REVEAL are blocked. The successor is disabled for global E013 discovery to preserve the enabled historical member without duplicates. Per-run parent PDFs and ZIPs are immutable; earlier output trees retain their run IDs.

References

[1] H. K. Moffatt, "The degree of knottedness of tangled vortex lines," *J. Fluid Mech.* 35, 117–129 (1969). DOI: 10.1017/S0022112069000991.

[2] P. G. Saffman, *Vortex Dynamics*. Cambridge University Press (1992). DOI: 10.1017/CBO9780511624063.

[3] R. L. Bishop, "There is More than One Way to Frame a Curve," *Amer. Math. Monthly* 82, 246–251 (1975). DOI: 10.1080/00029890.1975.11993807.